

Ultrasound imaging at 1.8 MHz is used to evaluate liver disease. The ultrasound signal travels through a boundary between liver and fat tissue at an angle of 30° relative to the normal at the boundary. If the ultrasound signal has a wavelength of 0.8 mm in fat tissue, what is the speed of sound in fat?

- ☐ A. 720 m/s [10%]
- ☐ B. 1250 m/s [18%]
- ✓ ☒ C. 1440 m/s [61%]
- ☐ D. 1660 m/s [9%]

Correct

61% answered correctly

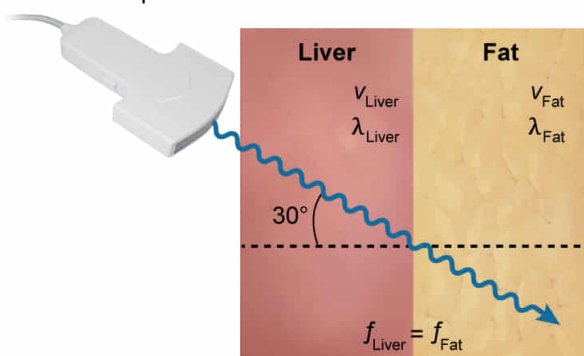
Explanation:

Calculating speed of sound in fat

$$v = \lambda f$$

- v Speed of sound
- λ Wavelength
- f Frequency

Ultrasound probe



In fat tissue:

$$\lambda = 0.8 \text{ mm} = 8 \times 10^{-4} \text{ m}$$

$$f = 1.8 \text{ MHz} = 1.8 \times 10^6 \frac{1}{\text{s}}$$

$$v = (8 \times 10^{-4} \text{ m}) \left(1.8 \times 10^6 \frac{1}{\text{s}} \right) = 1440 \frac{\text{m}}{\text{s}}$$

Ultrasound imaging utilizes sound waves at **frequencies** above the limit of human hearing to image inside the body. The **speed of sound** v changes in different mediums (eg, different tissues in the body) due to variations in the physical properties of the medium, but the frequency f of the sound wave remains constant. For a particular tissue, v equals the product of the **wavelength** λ and f :

$$v = \lambda f$$

In this question, λ is 0.8 mm and f is 1.8 MHz for the ultrasound signal in fat tissue. Substituting these values into the equation above yields:

$$v = (0.8 \text{ mm})(1.8 \text{ MHz})$$

Converting the units of λ to m and the units of f to Hz results in:

$$v = \left[(0.8 \text{ mm}) \left(10^{-3} \frac{\text{m}}{\text{mm}} \right) \right] \left[(1.8 \text{ MHz}) \left(10^6 \frac{\text{Hz}}{\text{MHz}} \right) \right]$$

$$v = (8 \times 10^{-4} \text{ m}) (1.8 \times 10^6 \text{ Hz})$$

$$v = 1440 \text{ m} \cdot \text{Hz}$$

The unit of Hz is equal to 1/s, so $\text{m} \cdot \text{Hz}$ is equivalent to m/s:

$$v = 1440 \frac{\text{m}}{\text{s}}$$

(Choice A) The value of 720 m/s is obtained by incorrectly multiplying by the sine of 30° .

(Choice B) The value of 1250 m/s is obtained by incorrectly

multiplying by the cosine of 30° .

(Choice D) The value of 1660 m/s is obtained by incorrectly dividing by the cosine of 30° .

Educational objective:

The speed of sound is equal to the product of the wavelength and frequency of the sound waves. When sound travels through different mediums, its speed and wavelength may change, but the frequency remains constant.

Time spent:0 sec

Subject:
Physics

QID:403472

System:
Light and Sound